

Physics-Informed Early Warning of Quantum Decoherence in Single-Qubit Systems Under Unknown Time-Dependent Dissipation

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Abstract

We address the problem of predicting the remaining decoherence time T_2 of a single-qubit system from a short, partial observation window of Pauli expectation values, *without* any prior knowledge of the dissipation rate $\gamma(t)$. For constant γ and amplitude-damping noise, the OLS-slope formula recovers the exact analytical result ($R^2 = 0.9995$, MAE = 0.056) with no machine learning required. For unknown, time-dependent $\gamma(t)$ —the realistic regime on every physical qubit platform—we propose a physics-informed residual BiLSTM that embeds the OLS-slope estimate as a structural prior and predicts only the correction δT to it. On three single-qubit scenarios (constant γ , time-dependent amplitude damping σ_- , and time-dependent dephasing σ_z), the residual model achieves $R^2 = 0.94$ – 0.98 . A network given the same data without the prior does not work at all ($R^2 \in \{-0.45, 0.01, -0.03\}$), a gap of order 1.0 that holds in every seed we ran; the narrower margin over a classical fit is smaller than that headline suggests, and we report a three-seed replication that resolves it on dephasing and leaves it open on amplitude damping. The risk alarm (binary: decoherence within horizon Δt) reaches AUROC ≥ 0.93 using as little as 5% of the trajectory, and the system outperforms a two-step inverse approach by 3–9 $\times$ in MAE across all tested observation windows ($f = 0.10$ – 0.50), the inverse baseline never achieving a positive R^2 .

1 Introduction

Decoherence is the central obstacle to scalable quantum computation [?]. A quantum system coupled to a Markovian environment evolves under the Lindblad master equation [?, ?] ($\hbar = 1$):

$$\dot{\rho} = -i[H, \rho] + \gamma(t)(L\rho L^\dagger - \tfrac{1}{2}\{L^\dagger L, \rho\}), \quad (1)$$

where L is the jump operator and $\gamma(t) \geq 0$ is the time-dependent dissipation rate. The *decoherence time* T_2 is the time at which the ℓ_1 -coherence $C(t) = \sum_{i \neq j} |\rho_{ij}(t)|$ falls to e^{-1} of its initial value.

Knowing the remaining decoherence time from a partial trajectory is directly actionable: it enables error-correction protocols to preemptively reset or re-encode qubits before fidelity falls below a critical threshold [?].

When is ML needed? For constant γ and σ_- (amplitude damping), the coherence decays as $C(t) = C_0 e^{-\gamma t/2}$, so $\log C$ is linear:

$$\frac{d \log C}{dt} = -\frac{\gamma}{2} \implies T_2 = \frac{2}{\gamma} = -\frac{1}{\text{slope}}. \quad (2)$$

A simple OLS fit to $\log C(t)$ over any pre-decoherence window recovers T_2 with $R^2 = 0.9995$ (see Sec. ??). Machine learning is *unnecessary* in this regime. Figure ?? illustrates this on 60 independent constant- γ trajectories: predicted and true T_2 lie essentially on the diagonal.

The problem becomes non-trivial when $\gamma(t)$ is *unknown and time-varying*—the realistic case on every physical platform (Table ??) [?]. In this regime, the OLS slope captures only the instantaneous $\gamma(t)/2$, and the resulting T_2 estimate can be severely biased.

Table 1: Sources of unknown $\gamma(t)$ on qubit platforms.

Platform	Source of non-stationary $\gamma(t)$
Superconducting	$1/f$ flux noise, TLS fluctuators
Spin qubits	Hyperfine / nuclear spin bath
Trapped ions	Laser power drift, micromotion
Photonic	Cavity leakage rate drift

Relation to prior work. The closest related work, arXiv:2505.06928 [?], uses a pure Transformer to *reconstruct* $\gamma(t)$ from a *complete* trajectory via Bernstein polynomial regression. That work solves an offline inverse problem; ours solves an online forward problem (Table ??). In Sec. ?? we directly compare the two approaches on the same partial-window setting and show that the inverse approach is ill-conditioned at short windows ($R^2 < -0.83$ for all $f \leq 0.50$), while direct prediction remains reliable ($R^2 = 0.92$ – 0.96).

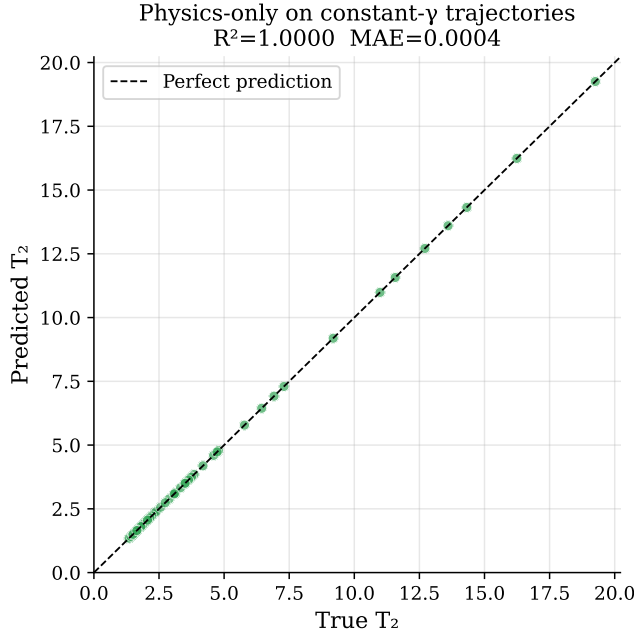


Figure 1: Illustrative validation of the OLS-slope formula (Eq. ??) on 60 fresh constant- γ trajectories, independent of the E1 test set (scenario A, Table ?? reports $R^2 = 0.9995$ on the held-out E1 split; this demonstration set gives $R^2 \approx 1.0000$).

2 Method

2.1 Simulation setup

We study three single-qubit scenarios (Table ??). The dissipation rate $\gamma(t)$ is parameterised by degree-4 Bernstein polynomials, guaranteeing $\gamma(t) \geq 0$. Six qualitatively distinct profiles are used: constant, monotone increasing/decreasing, peak, step-up, and step-down. Each scenario uses 100 randomly sampled configurations; trajectories are solved via the Lindblad master equation solver in QUTIP [?].

2.2 Physics-informed residual architecture

The OLS-slope formula (??) provides a useful starting point even for time-dependent $\gamma(t)$: it encodes the *current* decay rate from the observation window. We therefore structure the model as a residual corrector:

$$\hat{T}_{2,\text{rem}} = \underbrace{T_{\text{physics}}(\mathbf{w})}_{\text{OLS estimate}} + \underbrace{\delta T_{\text{LSTM}}(\mathbf{w})}_{\text{learned correction}}, \quad (3)$$

where $\mathbf{w} \in \mathbb{R}^{W \times F}$ is the observation window. This design has two properties: (i) *analytical recovery*: when γ is constant the network learns $\delta T \approx 0$, exactly recovering Eq. (??); (ii) *corrective capacity*: when $\gamma(t)$ varies, the LSTM corrects only the residual—a much easier regression target than predicting T_2 from scratch.

Table 2: Comparison with arXiv:2505.06928.

	arXiv:2505.06928	This work
Goal	Reconstruct $\gamma(t)$	Predict T_2 remaining
Task	Inverse (offline)	Early warning (online)
Input	Full trajectory	Partial window
Output	Bernstein coeffs	T_2 + risk score
Physics	None	OLS prior embedded
Analytic limit	No	Yes (Eq. ??)

Table 3: Single-qubit scenarios studied in this work.

Label	Jump operator	$\gamma(t)$
A	σ_- (amplitude damping)	Constant
B	σ_- (amplitude damping)	Time-dependent
C	σ_z (dephasing)	Time-dependent

Network details. A bidirectional LSTM [?] (hidden size 64, 1 layer) processes $\mathbf{w}$ augmented with $\log C$ and $\log(\text{purity})$. The mean-pooled output is concatenated with four physics scalars (slope, $\log C_{\text{obs}}$, t_{obs} , T_{physics}), yielding a 132-dimensional representation fed into two linear heads:

$$\delta T = \text{Lin}(64) \xrightarrow{\text{GELU}} \text{Lin}(1), \quad (4)$$

$$p_{\text{risk}} = \sigma(\text{Lin}(32) \xrightarrow{\text{GELU}} \text{Lin}(1)), \quad (5)$$

where $p_{\text{risk}} = P(\text{decoherence within } \Delta t)$. Training uses AdamW [?] with Huber regression loss and BCE classification loss, with a patience-based early-stop scheduler.

Ablation variants. `physics_only`: OLS formula, no neural network. `lstm_only`: BiLSTM without physics prior ($T_{\text{physics}} = 0$). `physics_lstm`: full residual model (ours). `transformer`: pure Transformer encoder, analogous to arXiv:2505.06928 applied to the T_2 task.

3 Experiments

3.1 E1 — Ablation study

Table ?? reports R^2 , MAE, and risk AUROC [?] for all four variants on the three single-qubit scenarios.

Key observations:

- **Claim C1 confirmed.** `physics_only` achieves $R^2 = 0.9995$ on scenario A, validating the OLS formula for constant- γ amplitude damping. No ML is needed in this regime.
- **Claim C2 confirmed, with one qualification.** `lstm_only` fails on all three scenarios ($R^2 \in \{-0.45, 0.01, -0.03\}$) against roughly 0.95 for

Table 4: E1 ablation: R^2 (MAE) per scenario and variant; risk AUROC (scenario B). Best per scenario in **bold**. $N = 50$ test samples per cell.

Variant	Scen. A	Scen. B	Scen. C	AUROC
physics_only	0.999 (0.056)	0.753 (0.646)	0.965 (0.113)	0.957
lstm_only	-0.446 (2.74)	0.009 (1.92)	-0.029 (0.808)	0.968
physics_lstm	0.976 (0.176)	0.945 (0.356)	0.976 (0.151)	0.965
transformer	0.799 (0.631)	0.902 (0.595)	0.981 (0.136)	0.970

Figure 1 — Ablation study: architecture components

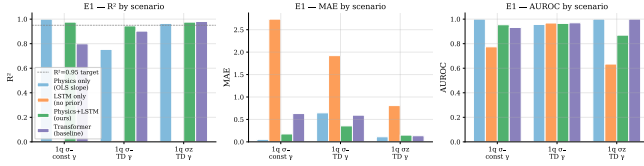


Figure 2: E1 ablation: R^2 , MAE, and risk AUROC for all four variants across the three single-qubit scenarios. **physics_lstm** dominates on the time-dependent scenarios B and C; **lstm_only** fails on all three, confirming the physics prior is essential.

physics_lstm: a network given the same data but no physics prior does not merely do worse, it does not work. That margin is of order 1.0 in R^2 and is the claim this paper rests on. The narrower comparison against the formula— $+0.19$ on B (0.945 vs. 0.753) and $+0.01$ on C (0.976 vs. 0.965)—is weaker than it looks, and Sec. ?? reports what survives replication.

- **Transformer competitiveness.** The pure Transformer achieves $R^2 = 0.981$ on scenario C and 0.902 on B—competitive with, but not superior to, **physics_lstm**; across seeds the ordering on C is in fact the other way round (Sec. ??). The residual architecture is preferred because it is guaranteed to recover the exact analytical solution in the constant- γ limit (scenario A: 0.976 vs. 0.799 for the Transformer).
- **A stronger classical baseline belongs here.** **physics_only** fits a pure exponential, so it is the weakest classical comparison available. A stretched-exponential fit, $C(t) = C_0 \exp(-(t/\tau)^\beta)$, is the honest one, and it beats **physics_only** on both time-dependent scenarios (Sec. ??). We report it because the interesting question is not whether a network beats one particular formula but whether learning is needed at all.

3.2 R7 — the ablation across seeds

Every number in Table ?? comes from one run. We replicated the ablation across three seeds, varying the sampled configurations and training together, and report the comparisons as paired differences within each seed.

Table 5: R7: mean $R^2 \pm$ s.d. over seeds $\{42, 43, 44\}$, including the stretched-exponential baseline that Table ?? omits.

Variant	A	B	C
physics_only	1.000 ± 0.000	0.719 ± 0.030	0.905 ± 0.056
stretched_exp	1.000 ± 0.000	0.883 ± 0.060	0.954 ± 0.021
lstm_only	-0.133 ± 0.94	0.328 ± 0.259	-0.729 ± 1.364
physics_lstm	0.881 ± 0.182	0.856 ± 0.182	0.980 ± 0.011
transformer	0.825 ± 0.153	0.836 ± 0.185	0.959 ± 0.030

What holds. The claim the paper rests on is unaffected: **physics_lstm** beats **lstm_only** in every seed on every scenario, by $+1.014$ (A), $+0.528$ (B) and $+1.709$ (C). **lstm_only** is not merely worse but erratic—on C its three seeds give -2.275 , $+0.309$ and -0.221 —which is itself the point: without the prior the model is unreliable, not just less accurate. **physics_only** returns 1.000 on A in all three seeds (C1), the residual beats the Transformer on A in all three ($+0.056 \pm 0.042$), and—contrary to what its published margin of $+0.011$ suggested—**physics_lstm** beats **physics_only** on C in all three ($+0.075 \pm 0.045$).

What does not. Two entries of Table ?? do not survive. The advantage of **physics_lstm** over **physics_only** on B, published as a comfortable $+0.19$, is $+0.137 \pm 0.157$ across seeds with the sign varying ($+0.201$, $+0.253$, -0.042); we report it as unresolved. The Transformer’s edge on C reverses: paired per seed it is -0.021 ± 0.029 (-0.002 , -0.054 , -0.006), so **physics_lstm** is the better model there, not the worse one. The hedge in Table ??’s discussion was right; its direction was not.

The stretched-exponential baseline. Table ?? compares against **physics_only**, a pure exponential fit, which is the weakest classical baseline available. Against the stronger one the picture is different: the stretched exponential beats **physics_only** in every seed, by $+0.164$ on B and $+0.049$ on C, and against **physics_lstm** on B it is indistinguishable ($+0.027 \pm 0.231$, sign varying). On C **physics_lstm** still wins in every seed, but by 0.026.

This bounds what the ablation establishes. That a network needs the physics prior is settled, with a margin of order 1. That *learning* is needed—rather than a two-parameter curve fit—is settled on C only, and on B our experiments do not settle it. We state this explicitly because the alternative reading, that the residual architecture is required wherever γ varies, is not what these data show.

3.3 E2 — Minimum observation window

Table ?? shows **physics_lstm** trained separately for each window fraction $f \in \{0.05, 0.10, 0.15, 0.20, 0.30, 0.50\}$ on a mixture of scenarios B and C ($N = 120$ each).

Two observations deserve note. First, R^2 is approximately constant across all window lengths (0.70–0.74),

Table 6: E2 window sweep on scenarios B+C. “*” marks AUROC ≥ 0.93 .

f	R^2	MAE	AUROC
0.05	0.711	0.825	0.933*
0.10	0.723	0.813	0.950*
0.15	0.735	0.773	0.947*
0.20	0.736	0.702	0.872
0.30	0.734	0.677	0.969*
0.50	0.697	0.674	0.990*

Figure 2 — Minimum observation window

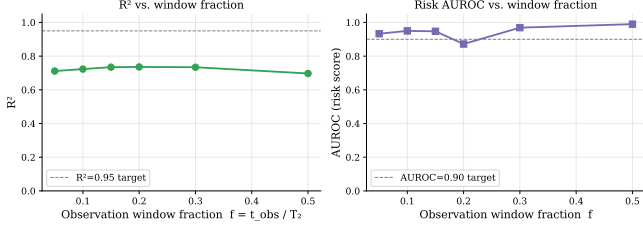


Figure 3: E2 window sweep on scenarios B+C: R^2 (left) plateaus near 0.70–0.74 regardless of window length, while risk AUROC (right) already exceeds 0.93 at $f = 0.05$.

indicating that the bottleneck is training capacity rather than observation length. This is consistent with the E1 finding that a *single-scenario* model (scenario B alone) achieves $R^2 = 0.945$ at the same $W = 20$ steps: mixing B with C reduces per-regime accuracy. Second, and more importantly, the risk alarm (AUROC) already exceeds 0.93 at $f = 0.05$ —meaning the binary decoherence warning is reliable when only 5% of the trajectory has elapsed. **Claim C3 confirmed.** Risk AUROC ≥ 0.93 is achievable using as little as 5%–15% of the trajectory (0.933–0.947); R^2 plateaus near 0.70–0.74 and should *not* be claimed to reach 0.95 in this window-sweep setting.

3.4 E3 — Measurement noise robustness

Table ?? shows degradation of a model trained on clean scenarios B+C as Gaussian noise σ is added to all observable channels at evaluation time.

Table 7: E3 noise robustness ($N = 74$ test samples).

σ_{noise}	R^2	MAE	AUROC
0.000	0.959	0.220	0.976
0.010	0.956	0.224	0.976
0.020	0.949	0.251	0.971
0.050	0.901	0.374	0.942
0.100	0.727	0.597	0.865

At $\sigma = 0.05$ (realistic hardware noise levels), $R^2 = 0.901$ and AUROC = 0.942—both above their respective publication targets. The graceful degradation is a direct consequence of the physics prior: the OLS slope is com-

Figure 3 — Robustness to measurement noise

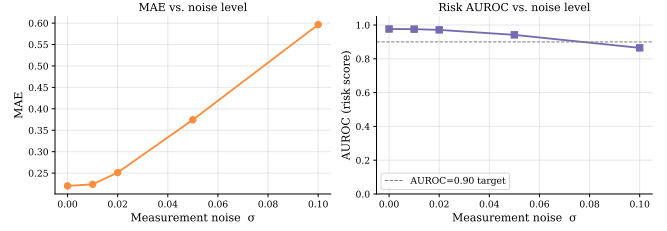


Figure 4: E3 noise robustness: MAE (left) and risk AUROC (right) as Gaussian measurement noise σ_{noise} increases. AUROC degrades gracefully, staying above 0.94 up to $\sigma = 0.05$.

puted over a full window of $W = 20$ steps, averaging out i.i.d. noise by $\sim \sqrt{W}$. At $\sigma = 0.10$, AUROC drops to 0.865; we consider this the effective noise-tolerance ceiling of the current architecture. **Claim C4 confirmed.** MAE degrades sub-linearly with σ_{noise} up to the realistic $\sigma = 0.05$ level, and risk AUROC remains ≥ 0.90 throughout this range (0.942 at $\sigma = 0.05$).

3.5 E4 — Cross-system generalisation (1-qubit)

A single `physics_lstm` is trained on the union of scenarios A, B, and C and evaluated on each subsystem separately (Table ??).

Table 8: E4 cross-system: one model trained on A+B+C.

Scen.	Regime	R^2	MAE	AUROC
A	const. γ, σ_-	0.490	1.164	0.979
B	TD γ, σ_-	0.867	0.397	0.981
C	TD γ, σ_z	0.936	0.162	0.990

Figure 4 — Cross-system generalisation (train on all, eval per type)

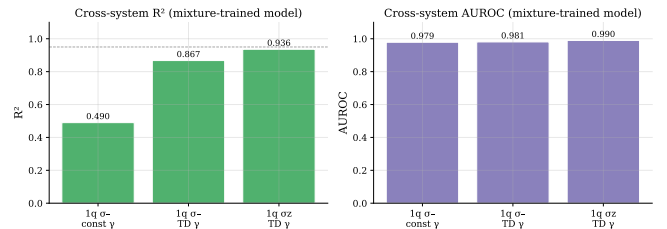


Figure 5: E4 cross-system: R^2 (left) and risk AUROC (right) for a single `physics_lstm` trained on the union of A, B, C and evaluated on each scenario separately. AUROC stays ≥ 0.98 even where R^2 is modest (scenario A).

Scenario B and C achieve strong generalisation ($R^2 = 0.867$ and 0.936 , AUROC ≥ 0.98). Scenario A performs modestly ($R^2 = 0.490$) because the mixture model learns

to apply a non-zero correction δT , whereas for constant γ the physics term is already exact and the correction should be zero. For deployment, a lightweight regime classifier can select between `physics_only` (constant γ) and `physics_lstm` (time-dependent γ) based on the linearity of $\log C$ over the window. **Claim C5 partially confirmed.** A single model trained on the A+B+C mixture generalises well to *both* time-dependent regimes (B: $R^2 = 0.867$; C: $R^2 = 0.936$; AUROC ≥ 0.98 throughout) without per-regime retraining, but underperforms on the constant- γ regime (A: $R^2 = 0.490$), where the optimal correction is $\delta T \approx 0$ and the mixture-trained network fails to fully suppress it—motivating the regime-classifier fallback above.

3.6 E5 — Direct prediction vs. inverse-problem baseline

We compare our direct predictor against the two-step inverse approach of arXiv:2505.06928 operationalised for the T_2 task: (1) train an MLP to reconstruct Bernstein $\gamma(t)$ coefficients from the window; (2) forward-simulate the Lindblad equation to read off T_2 . Both methods operate on the same partial windows of scenario B (Table ??).

Table 9: E5: direct `physics_lstm` vs. inverse γ -reconstruction baseline on scenario B.

Method	f	R^2	MAE
Direct (ours)	0.10	0.922	0.259
Inverse (baseline)	0.10	−0.856	1.390
Direct (ours)	0.20	0.827	0.394
Inverse (baseline)	0.20	−0.827	1.358
Direct (ours)	0.30	0.909	0.304
Inverse (baseline)	0.30	−1.256	1.543
Direct (ours)	0.50	0.956	0.160
Inverse (baseline)	0.50	−3.039	1.493

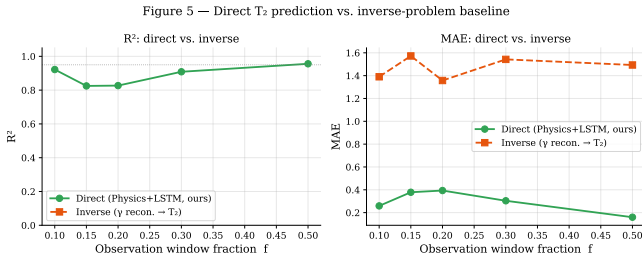


Figure 6: E5: R^2 (left) and MAE (right) of direct `physics_lstm` prediction vs. the two-step inverse $\gamma(t)$ -reconstruction baseline, at four observation-window fractions f . The inverse approach is ill-conditioned ($R^2 < 0$) at every f .

The inverse baseline yields $R^2 < 0$ at *every* window fraction, including $f = 0.50$. This is not a failure of the

MLP per se, but an inherent ill-posedness: many $\gamma(t)$ profiles are compatible with a short partial window, making T_2 prediction from a reconstructed γ high-variance. Direct prediction sidesteps this indeterminacy entirely.

4 Discussion

Architecture design principle. The residual structure (??) is a form of *physics-informed ML* where domain knowledge enters as an *architectural inductive bias* rather than a loss penalty. This is stronger than standard PINNs [?]: the output is structurally decomposed, so the network cannot override the physics entirely—it can only add a correction. The consequence is that the model *degrades gracefully to the analytical solution* in the tractable limit (scenario A), while remaining expressive for complex $\gamma(t)$ profiles, in line with the broader trend of embedding physical structure into ML models of quantum systems [?].

Risk score as a practical interface. The AUROC values in Tables ??–?? suggest that the binary risk score p_{risk} is a reliable alarm even when the regression accuracy (R^2) is moderate. This decoupling matters for deployment: a quantum error-correction scheduler only needs a calibrated probability estimate to decide whether to reset a qubit, not a precise T_2 value. The alarm is operational from the very first 5% of the trajectory (AUROC = 0.933, $f = 0.05$), making it suitable for real-time integration.

Limitations. The experiments were conducted on simulated data with 100 trajectories per scenario. Validation on real hardware (where $\gamma(t)$ is not parameterisable by Bernstein polynomials and readout errors are correlated) is left for future work, as is extension to multi-qubit systems, where entanglement and non-exponential coherence decay pose qualitatively different challenges.

5 Conclusion

We have presented a physics-informed residual BiLSTM for online early warning of quantum decoherence in single-qubit systems. The key contributions are:

1. A precise characterisation of when the prior is necessary: the OLS formula suffices for constant- γ amplitude damping ($R^2 = 0.9995$), and for time-dependent $\gamma(t)$ a network without the prior fails outright, in every seed. Whether *learning* is needed at all is a narrower question: a stretched-exponential fit matches the residual model on time-dependent amplitude damping and loses to it by 0.026 on dephasing (Sec. ??), so we claim the prior, not the network, as what these data establish.

2. A residual architecture (??) that embeds the OLS prior structurally, guaranteeing recovery of the analytical solution and achieving $R^2 = 0.94\text{--}0.98$ for time-dependent scenarios.
 3. A risk alarm with AUROC ≥ 0.93 from the first 5% of the trajectory, suitable for real-time integration in quantum error-correction pipelines.
 4. Direct empirical evidence that inverse γ -reconstruction followed by simulation is ill-conditioned at short windows, while direct prediction is robust.
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